

A Programmable-Oscillator-Based Frequency-Shift Keying Radio Frequency Transmitter for Wildlife Tracking

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Radio frequency (RF) telemetry is an important technology for tracking animals. RF tags (transmitters) need to be lightweight, energy efficient, and small to study a wide range of wildlife. Transmitters in such systems must modulate high chip-rate pseudorandom sequences for 3D localization, but commercial transceiver chips struggle to meet these coding requirements and physical constraints simultaneously. To solve this, programmable oscillators were used to engineer frequency-programmable tags utilizing on-off keying (OOK) modulation. However, many established tracking systems use frequency-shift keying (FSK) modulation. Here, a prototype design demonstrates successful FSK signal generation and existing receiver compatibility for a preliminary test distance of 342 m. Another design adds RF circuitry to test whether additional size, complexity, and power draw improved signal quality. Between the two designs, laboratory benchmarks show that there is no significant difference in transmission power at different supply voltages. Overall, the developed prototype and its firmware provide the foundation for FSK-enabled, miniaturized, and ultra-low power 3D-tracking wildlife tags.

I. INTRODUCTION

Since the 1960s, researchers have attached radio frequency (RF) tags to both terrestrial and aquatic animals to study their movement.¹ Today, these tags are critical to the study of ecology and informed development of sustainable energy and agriculture infrastructure.²⁻⁹ Many old tag designs use very high frequency (VHF) radio transmitters and a human receiver.¹⁰ To combat the dependence of human operators, some active tags incorporated GPS modules to transmit positional data directly. However, this significantly increased the energy consumption, size, and mass of the tag. This reduces tag lifespan, increases recapture frequency, and introduces bias in data by causing potential burdens on the animal.¹⁰ Thus, to circumvent GPS, researchers have developed passive tags for RF-telemetry based tracking systems such as time difference of arrival (TDOA).^{1,10-14}

By using these positioning systems, tags have become lighter, cheaper, and longer lasting.^{10,11} Though, as reduced tag mass encourages more researchers to study even smaller animals,¹⁵ the benefits of simply removing GPS do not last long.¹⁶ Researchers recommend that a tag should weigh less than 5% of a bird or bat's body mass.¹⁷ For instance, some species of *Myotis* typically weigh less than 5 g, which requires tags to be at most 0.25 g. Hence, focus shifted to optimizing either the transmitter design or battery charge density. Nevertheless, there is slow effort in improving transmitter power consumption and size due to the multiple components in VHF radio circuitry,¹⁸ including an oscillator, frequency synthesizer, modulator, and power amplifier.¹⁴

As hardware RF designs typically optimize for one frequency, it also makes supporting the multiple frequency bands allocated for wildlife tracking difficult. A different option is commercial RF transceivers, which combine all components into a compact, programmable chip. However, many chips that support on-off keying (OOK) and frequency-shift keying (FSK) fail to generate high chip-rate pseudorandom codes used in TDOA systems while simultaneously minimizing footprint size and power consumption.¹⁹⁻²²

PNNL researchers have developed TDOA tags that use programmable oscillators to solve this problem while outperforming commercial tracker options.^{23,24} These tags use OOK modulation of a 1 MHz chip-rate Gold code. However, large wildlife tracking networks that have proven success either use or support FSK tags.^{7,11} Adopting FSK would therefore allow existing infrastructure and its broad research community to easily integrate and use PNNL's programmable oscillator tags. Hence, this work repurposes the programmable oscillator tag architecture for FSK onto a prototype printed circuit board (PCB) and characterizes resulting performance to test novel systems and investigate potential tradeoffs.

II. BACKGROUND & THEORY

TDOA systems compare signal time-of-arrivals at different receivers to triangulate transmitter position. The tag must generate a unique RF waveform that can be parsed by the receivers for signal arrival time. To do this, tags transmit pseudorandom binary sequences such as Gold codes with attractive autocorrelation properties. For accurate time-of-arrival measurements, tags must be able to modulate high-chip rate Gold code onto synthesized carrier signals.

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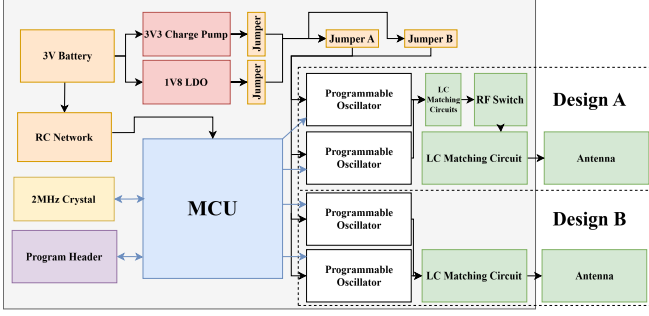


FIG. 1. Circuit diagram of PCB with multiple test designs.

PNNL engineers leveraged the Fourier series of a DC-offset square-wave oscillating between amplitudes A and 0 with frequency ω , as in Equation 1, to generate the carrier using a programmable square-wave oscillator.

$$f(t) = \frac{A}{2} + \frac{2A}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\sin(n\omega t)}{n}. \quad (1)$$

Engineers then programmed oscillators to either 165 MHz to use the first harmonic for high output power, or 55 MHz to generate a 165 MHz carrier using the third harmonic to balance service life and detection range. The OE (output/enable) pin of the oscillator was toggled repeatedly to modulate Gold code onto this carrier with OOK.¹⁸

Discontinuous FSK can be modeled as two OOK signals with two different carrier frequencies f_+ , $f_- = f_c \pm \Delta f$ that have a center frequency f_c and frequency deviation Δf . Applying this logic, two programmable oscillators set at f_- and f_+ can be enabled on and off for the respective binary symbol they represent.

III. DESIGN & IMPLEMENTATION

As this project investigates programmable oscillator architecture for FSK modulation instead of miniaturization, the designed PCB incorporates multiple configurable jumpers and RF circuitry designs, as seen in Figure 1.

Three voltage levels are provided to power the oscillators during testing using jumpers. The charge pump provides a 3.3 V supply, the low-dropout regulator (LDO) provides a 1.8 V supply, and one more jumper directly connects the 3 V battery to the oscillators. Two more jumpers provide the option to select which of two designs are powered during testing. The RC network that follows the battery input prevents microcontroller unit (MCU) brown outs.

This prototype uses an 8-bit MCU that has proven track record for prototype OOK modulation PCBs under the Bat Tech project at PNNL. An MCU GPIO pin directly powers a 2 MHz oscillator, since the oscillator's

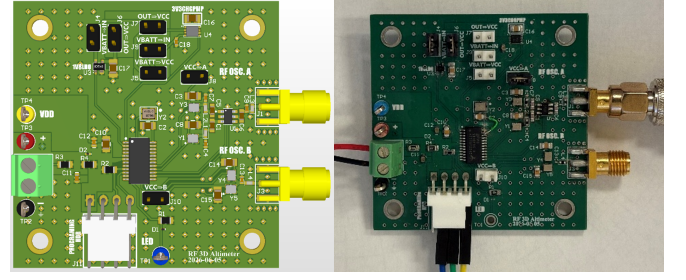


FIG. 2. PCB in design versus PCB after assembly.

maximum current draw of 3 mA is less than the 5 mA the MCU can provide. An alternative method that toggles the enable pin of the oscillator ignores the parasitic current draw it introduces. The 2 MHz oscillator is used to modulate Gold code at a 1 MHz chip-rate, since GPIO write operations take two clock cycles to complete. Assembly is used as firmware to achieve precise control over execution clock cycles, and Python is used to generate transmission Assembly code from a 2047-bit Gold code sequence.

Both RF circuitry designs have two programmed oscillators at f_- and f_+ and include GPIO pins that are connected to each oscillator's OE pin. When the OE pin is driven low, its output pin is pulled down with 500 k Ω and the output driver is disabled. When the OE pin is driven high, a square wave of the programmed frequency is outputted.

A rudimentary implementation of FSK shorts two oscillator outputs together to the same node. However, internal reflections may significantly deteriorate signal quality and damage the oscillators, so the implementation of two designs investigates this potential.

Design A implements an RF switch to isolate and toggle between the two oscillators' outputs. Design B connects the oscillators' outputs into one node. For design A, the RF switch has ports that are matched to 50 Ω impedance and require 1000 pF DC blocking capacitors. Since previous measurements suggest an oscillator Thevenin output resistance of 200 Ω at the desired frequency range,²⁴ additional footprints for LC matching networks are added. Specifically, from the oscillators to the RF switch ports, shunt capacitors are placed before series inductors to match to a lower impedance. Moreover, both designs include similar LC match networks between output ports to SMA connector footprints. 0 Ω 0402 resistors can be soldered in place of series inductors to bypass the LC matching circuits.

The modulation index for FSK is defined as

$$h = \frac{2\Delta f}{R_c}, \quad (2)$$

where R_c is the Gold code chip-rate and Δf is the deviation from center frequency f_c . The desired chip-rate and center frequency is 1 MHz and 161.325 MHz, respectively. Therefore, $\Delta f = 500$ kHz, $f_- = 160.825$ MHz, and

TABLE I. Test configurations.

Configuration	Design	V_{CC}
C1	A (RF switch)	3.3 V
C2	A 20	1.8 V
C3	B (No RF switch)	3.3 V
C4	B	1.8 V

$f_+ = 161.825$ MHz. Two oscillators were programmed to use the first harmonic to generate both f_- and f_+ frequencies for each design.

To ensure good signal quality, coplanar waveguides are matched to $50\ \Omega$ for all RF traces. Ground stitching vias are placed in a grid pattern with 2.5 mm spacing. RF signals are kept as short as possible to simulate hypothetical dimension constraints on miniaturized tags and reduce internal reflections. Power planes are also used.

Initial free-space propagation tests used 31.43 mm pre-made wire antennas. The center signal pad of the SMA connector is used to solder the wire antenna on. Later long-range, free-space propagation tests used omnidirectional antennas. Solder blobs may reduce signal quality, but two possible functions in one footprint provide greater research flexibility.

IV. RESULTS & DISCUSSION

Four design configurations are listed in Table I. V_{CC} denotes the oscillator supply voltage. A SA44B spectrum analyzer and DSO7014B oscilloscope were used to analyze the signals generated by each configuration. Current draw was measured with a shunt resistor between the system and power supply. Lastly, a T3VNA200 was used to find passive circuit impedance and match characteristics.

For each of the four configurations, there are three continuous wave (CW) transmission tests: subtest 1 (CW-), subtest 2 (CW+), and subtest 3 (CW+-). Subtest 1 toggles only the oscillator responsible for f_- , the subtest 2 toggles only the oscillator responsible for f_+ , and subtest 3 toggles between the two frequencies to simulate continuous FSK transmission.

One more test remodified old OOK tag firmware from PNNL's Bat Tag Project for FSK. This firmware included a sleep-and-transmit duty cycle. Power draw and lifespan calculations are derived from measurements of a prototype running this firmware. This test is referred to as "life span test".

For all tests, design A and B used $0\ \Omega$ resistors to bypass the LC matching circuits. "Power" refers to the integrated power within a 20 MHz bandwidth centered on the first-harmonic peaks, while "99%-bandwidth" refers to the bandwidth, within a 200 MHz span using the same center frequency, that contains 99% of the measured spectral power. Across all tests, the power under the 20 MHz bandwidth was higher than the power under the 99%-bandwidth by a mean of (0.184 ± 0.140) dBm, where

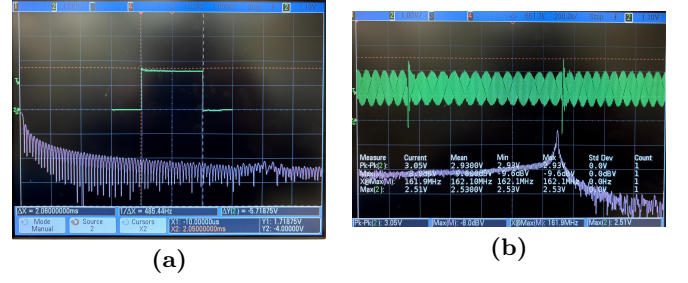


FIG. 3. Captured signal from antenna pad for configuration C3 in the time domain. (a) Gold code burst at 1 ms/div. (b) Zoomed-in segment with frequency toggles and phase discontinuities at 200 ns/div.

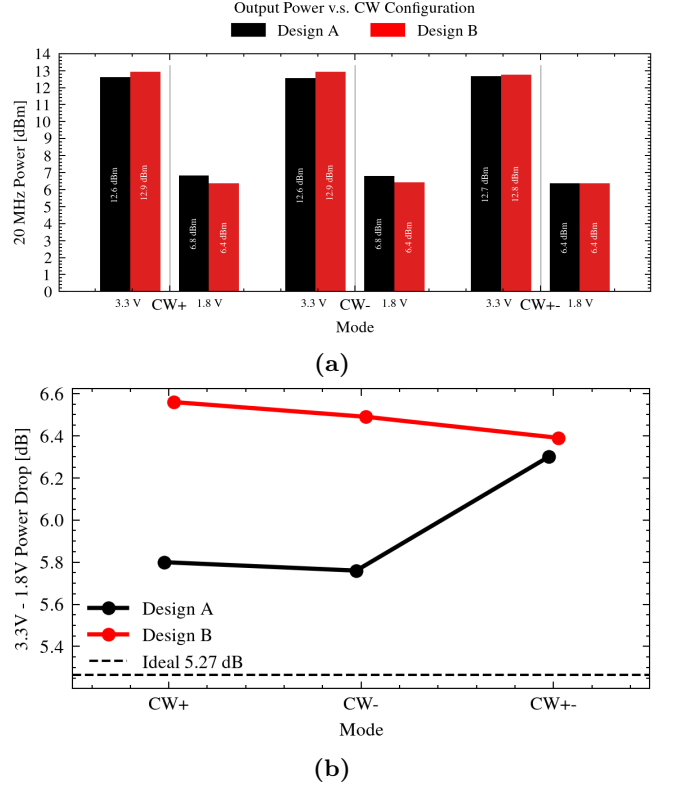


FIG. 4. (a) Output power across a $50\ \Omega$ load for test configurations. (b) Difference in output powers for V_{CC} drop.

uncertainty represents one standard deviation across test configurations.

A. Signal Characteristics

Under the life span test, all oscillators generated signals with frequency f_- or f_+ at a 1 MHz chip-rate with no overlap. As shown in Figure 3 (b), clear phase discontinuities occurred for every transition between the two carrier frequencies. Similarly, Figure 3 (a) shows each Gold code transmission lasting the intended 2 ms, repeating every 2 s.

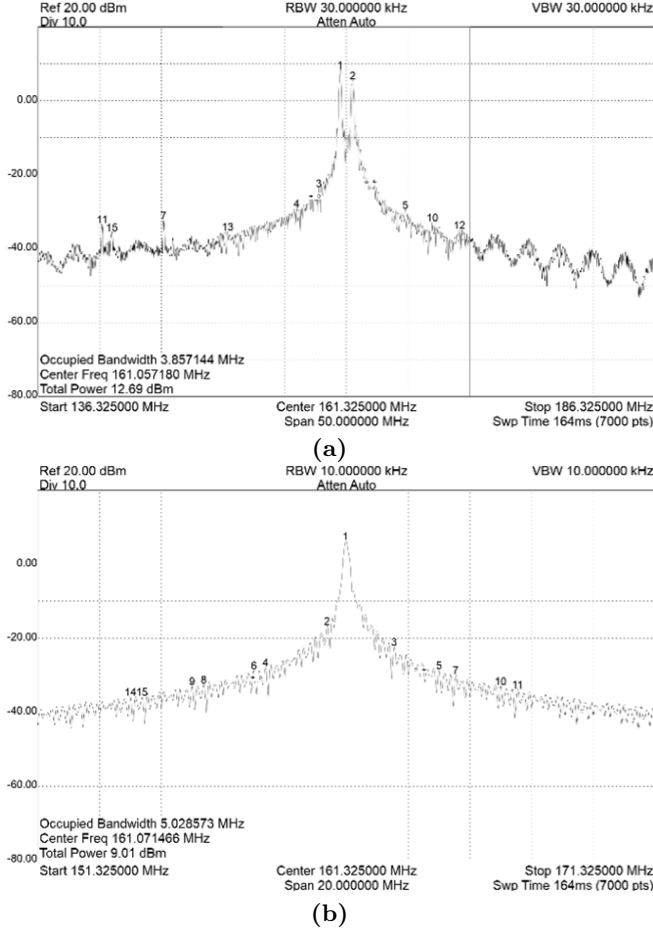


FIG. 5. (a): 50 MHz span spectrum plot of output from test C3. Peak 1 has a value of 8.32 dBm at 160.84 MHz. Peak 2 has a value of 5.33 dBm at 161.84 MHz (b): 50 MHz span spectrum plot of CW OOK at 3.3 V. Peak 1 has value 6.28 dBm at frequency 161.32 MHz.

Seen in Figure 4 (a), at $V_{CC} = 3.3$ V, average power is (12.70 ± 0.15) dBm. At $V_{CC} = 1.8$ V, average power drops to (6.34 ± 0.21) dBm. In Figure 4 (b), the total average power difference for a voltage drop from 3.3 V to 1.8 V is 6.21 dB, though the expected value is 5.27 dB. Additional loss is likely due to voltage-dependent CMOS output resistance.

Spectrum plots for all combinations have two distinguishable tones at f_- and f_+ , as seen in Figure 5 (a). The measured frequency for f_- is 160.84 MHz and f_+ is 161.84 MHz; both have a 15 kHz frequency difference from their respective programmed frequencies.

For subtest 3 transmissions, the 99%-bandwidth is 3.78 MHz. At 3.3 V, output peak-to-peak voltage is approximately 1.85 V. Moreover, the oscillator for f_- has a stronger output power than the oscillator for f_+ by 2.56 dB. For subtest 1 and 2 transmissions, the 99%-bandwidth is 114.29 kHz.

The relationship between design choice and measured output power is not significant. This is different from what is expected, since 0 Ω bypass resistors connect the

oscillators directly to 50 Ω ports of an RF switch in design A, introducing parasitic resistance and power loss. An explanation for these results may be the short trace lengths in all the RF circuitry as well as the 50 Ω trace impedance match.

For design B, the oscillator responsible for f_- became damaged after some initial testing. A replacement was re-soldered on, and the system worked without problem. It is suspected that internal reflections damaged the oscillator, since design B connects two output pins into one node.

An oscillator in an OOK tag prototype was benchmarked at 3.3 V. As seen from Figure 5 (b), the 99%-bandwidth contains 9.01 dBm of power. Compared to a system performing FSK modulation at 3.3 V in Figure 4, it is approximately 3 dB less. This is expected, as FSK has a constant envelope compared to the 50 % on-off duty cycle of OOK. Thus, for the same baud rate, FSK takes two times more power than OOK. This allows for longer range detections but reduces the lifespan of tags compared to OOK.

B. System Impedance Match & Output Resistance

Reflection coefficient magnitudes of the unpowered prototype were measured using port 1 of the VNA. At both f_- and f_+ , $|\Gamma| = 0.928$ for design A, and $|\Gamma| = 0.955$ for design B. So, relative to a 50 Ω reference impedance, both designs have a poor match. However, this result should not be interpreted as the impedance match of the powered oscillator, as unpowered measurements include parasitic capacitance from components which behave differently under power.

It is possible to estimate an equivalent output impedance by only considering the real resistance. The Thevenin resistance R_s of the oscillator circuits can be calculated with:

$$R_s = \left(\frac{\frac{2V_{CC}}{\pi\sqrt{2}}}{R_L \cdot \sqrt{10^{P/10}}} - 1 \right) R_L, \quad (3)$$

where $R_L = 50 \Omega$, $V_{CC} \in \{3.3 \text{ V}, 1.8 \text{ V}\}$, and P is the measured power of the first harmonic in dBm. This assumes that the waveform is an ideal square wave, the Thevenin source is linear, and that reactance is negligible.

From Equation 3, the average active circuit output resistance for all twelve test configuration and subtest combinations is estimated to be 31 Ω . However, the average output resistance of the circuit is $(27 \pm 1) \Omega$ for tests at $V_{CC} = 3.3$ V, while $(35 \pm 2) \Omega$ for tests at $V_{CC} = 1.8$ V.

C. Lifespan & Power Consumption

Table II contains life span comparisons. For life span tests, the battery voltage was 3.0 V, and transmission

TABLE II. Life span calculations for different supply voltages and operating harmonics.

System and V_{CC}	Energy per transmission	Service life for 2s ping rate
OOK 1.8 V 5 th Harmonic ^a	116 μ J	15 days
OOK 1.8 V 3 rd Harmonic ^a	149 μ J	12 days
FSK 3.3 V 1 st Harmonic	251 μ J	6 days
ATLAS FSK 3.3V ^b	737 μ J	2 days

^a Data from Ref. 18.^b Data from Ref. 25.

length was 2 ms. All transmissions were preceded by a short burst of power draw, likely due to the 2 MHz oscillator turning on and MCU waiting for lock. At $V_{CC} = 3.3$ V each short burst lasted 0.168 ms, and drew 41 mA, while each transmission drew on average 38 mA.

Service life can be calculated from battery capacity, transmission current draw, ping rate, and parasitic current.¹⁸ Since derivation is commonplace, only final values are given. Battery capacity is assumed to be 6.2 mAh and use measured maximum static current of 1.7 μ A. The battery capacity is derived from PNNL OOK tags for proper comparison between multiple systems.¹⁸

Results show that oscillator-based designs yield significantly less power consumption compared to a design that utilizes a commercial FSK transceiver in ATLAS. However, the FSK prototype consumes more energy per transmission than OOK designs. Some explanations include:

1. The first harmonic is used for carrier generation in the FSK prototype rather than the third or fifth in the OOK prototype.
2. The developed prototype utilizes a family of programmable oscillators that consume significantly more current.
3. Stray PCB components on the prototype increase static current draw.
4. FSK draws roughly two times more power than OOK.
5. The benchmarked FSK system boosts the battery voltage to 3.3 V, drawing more current than 1.8 V LDO systems.

Miniaturized designs of the engineered prototype with an LDO and the same oscillators as OOK tags using the third or fifth harmonic will inherently solve most problems listed.

D. Field Tests

Preliminary field tests for up to 50 m using both designs demonstrated successful detection using repurposed OOK receiver infrastructure and firmware by PNNL scientists. Additional field tests up to 342 m tested design

B and demonstrated successful detections as well. Previous long-range tests of OOK tags demonstrated more than 20 km of range; similar, if not better, performance is expected with FSK prototype PCBs.

V. CONCLUSION & NEXT STEPS

An RF transmitter prototype was developed that investigates and benchmarks FSK modulation using programmable oscillators. Successful characterization and field testing demonstrated FSK modulation at 1 MHz chip-rate using programmable oscillators. Future stress tests are recommended on design B to investigate the true failure mode of oscillator f_- and benchmark reliability. Since TDOA systems struggle to effectively resolve three-dimensional spaces, hardware and firmware design for a barometer is also recommended. Together, this prototype and study provide a critical steppingstone to developing low power, low mass, and small footprint animal tracking RF tags.

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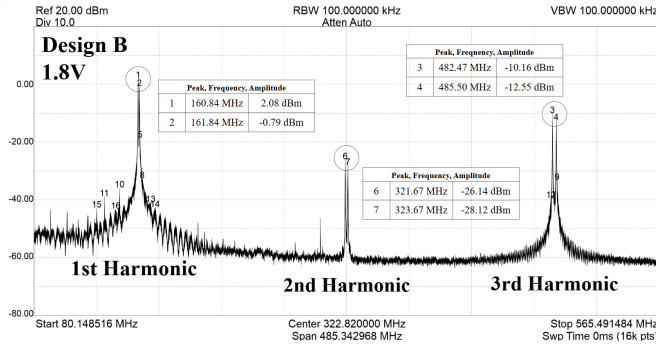


FIG. 6. Measured harmonics of outputted square wave from test configuration C3 and subtest 3.

Appendix A: Square-Wave Harmonic Content

Figure 6 shows that current revisions of the prototype PCBs transmit multiple harmonics of the square-wave output signal. A bandpass filter could be implemented in future revisions to reduce the total emission bandwidth and select the intended harmonic for transmission.

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